IGCSE 0607 Practice Questions — E7.4 — Combined Questions & Markscheme

Trigonometric functions

Total Questions: 12 — **Total Marks:** 63 Marks

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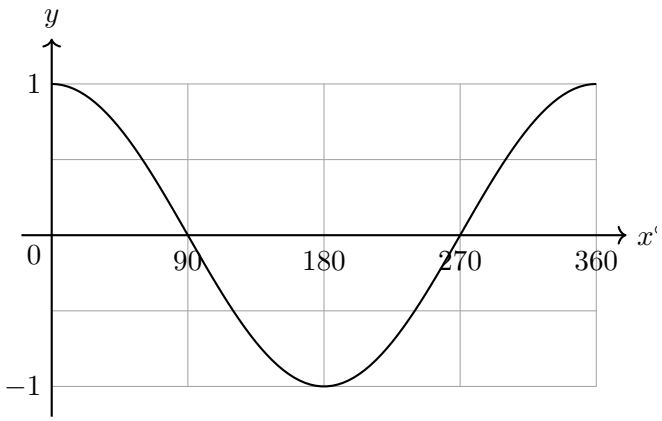
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1 Solve the equation for $0° \leq x \leq 360°$.

$\sqrt{2}\sin x - 1 = 0$

$x =$ ..... **[3]**

2 The diagram shows the graph of $y = \cos x$ for $0° \leq x \leq 360°$.



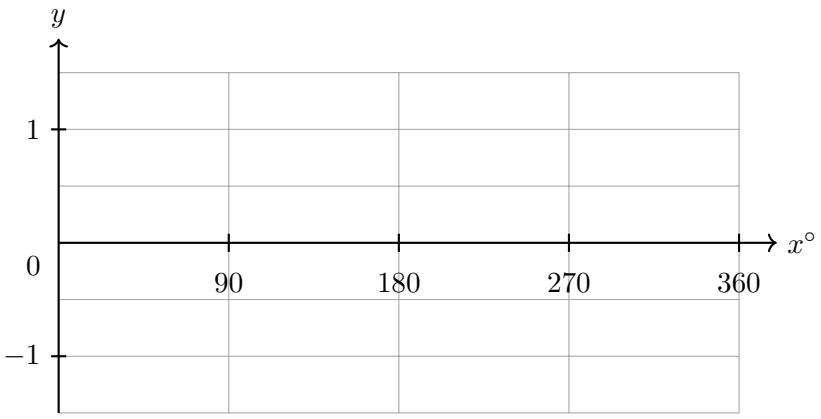
(a) Write down the coordinates of the lowest point on the graph.

(..... , ...) **[1]**

(b) Use the graph, or otherwise, to solve the equation $\cos x = 0.5$ for $0° \leq x \leq 360°$.

$x =$ ..... **[2]**

3 **(a)** On the grid, sketch the graph of $y = \sin x$ for $0° \leq x \leq 360°$.



[2]

(b) Write down the exact solutions to the equation $\sin x = 0.5$ for $0° \leq x \leq 360°$.

$x =$ ..... , $x =$ ..... **[2]**

4 Answer the following questions for $0° \leq x \leq 360°$.

(a) Solve the equation

$$2\cos x - 1 = 0$$

$x =$ ..... **[3]**

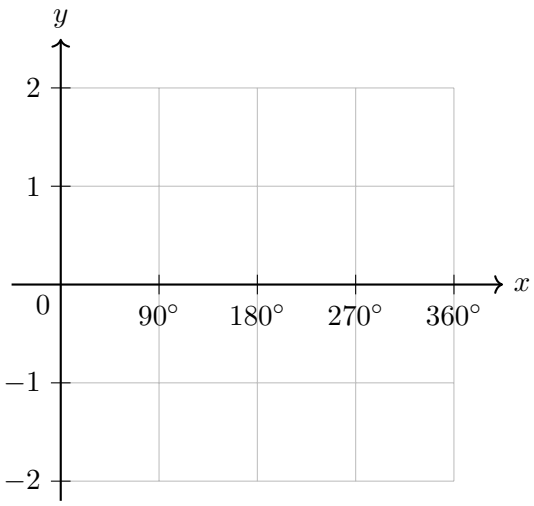
(b) Write down the number of solutions to the equation

$$3\sin x = 5$$

..... **[1]**

5 The properties of trigonometric graphs can be used to solve equations.

(a) On the grid, sketch the graph of $y = \cos x$ for $0° \leq x \leq 360°$.



[2]

(b) Write down the coordinates of the minimum point on this graph.

(..... , ...) **[1]**

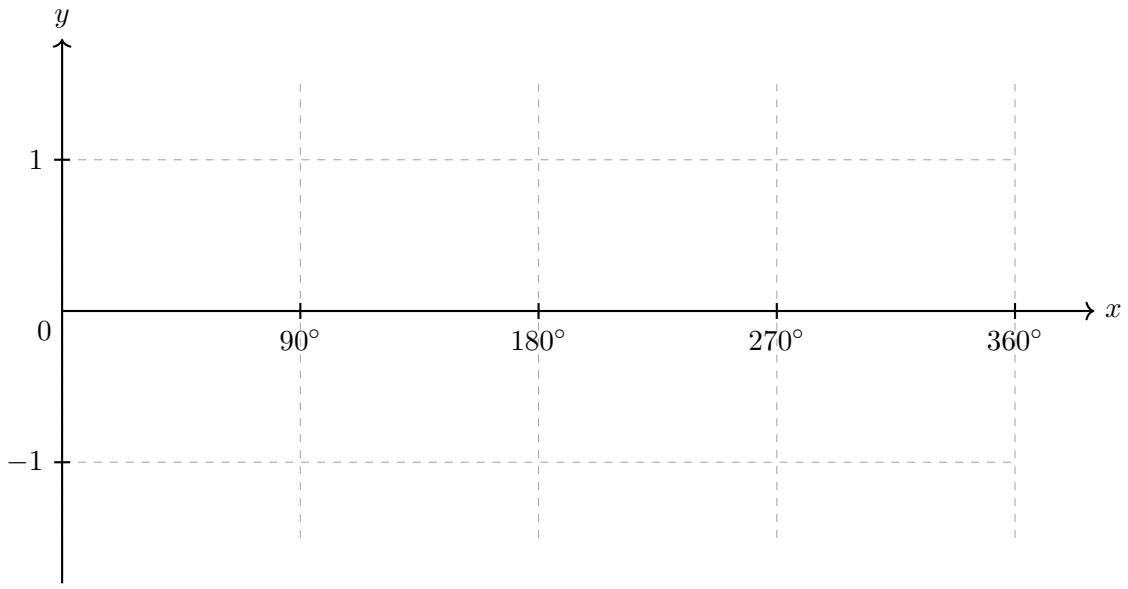
(c) Solve the equation for $0° \leq x \leq 360°$.

$2\cos x + 1 = 0$

$x =$ ..... **[3]**

6 The functions $g(x) = \sin x$ and $h(x) = \cos x$ are defined for $0° \leq x \leq 360°$.

(a) On the grid provided, sketch the graphs of $y = g(x)$ and $y = h(x)$.



[3]

(b) Solve the equation $g(x) = h(x)$ for $0° \leq x \leq 360°$.

$x =$ ..... **[2]**

(c) Use your graphs to write down the range of values of x for which $g(x) > h(x)$ for $0° \leq x \leq 360°$.

..... **[2]**

7 Solve the following equations for $0° \leq x \leq 360°$.

(a) $2\sin x + \sqrt{3} = 0$

$x =$ ..... **[3]**

(b) $4\cos^2 x - 3 = 0$

$x =$ ..... **[4]**

8 Consider the trigonometric function $f(x) = 4\sin x + 3$ for $0° \leq x \leq 360°$.

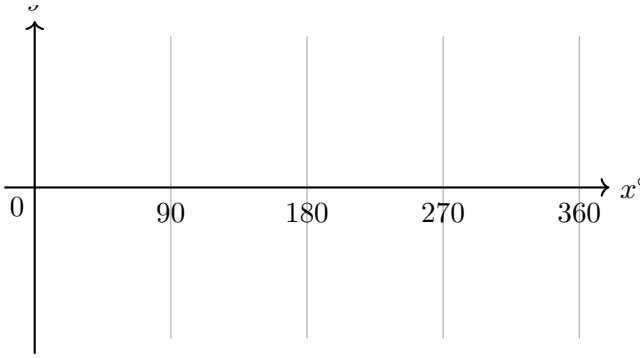
(a) Solve the equation $f(x) = 0$.
Give your answers correct to 1 decimal place.

$x =$ ..... **[3]**

(b) Write down the exact coordinates of the minimum point on the graph of $y = f(x)$.

(..... , ...) **[2]**

9 **(a)** On the axes provided, sketch the graph of $y = \tan x$ for $0° \leq x \leq 360°$.



[2]

(b) Solve the equation $3\tan x + 5 = 0$ for $0° \leq x \leq 360°$.
Give your answers correct to 1 decimal place.

$x =$ ..... **[3]**

10 **(a)** Solve the equation for $0° \leq x \leq 360°$.

$$4\cos x + 1 = 0$$

Give your answers correct to 1 decimal place.

$x =$ ..... , $x =$ ..... **[3]**

(b) Solve the equation for $0° \leq x \leq 360°$.

$$2\tan x - 5 = 0$$

Give your answers correct to 1 decimal place.

$x =$ ..... , $x =$ ..... **[3]**

11 **(a)** The equation $a\sin x - b = 0$ has a solution $x = 30°$.
Given that a and b are positive integers with no common factors, find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[2]**

(b) Write down the second solution to the equation $a\sin x - b = 0$ for $0° \leq x \leq 360°$.

$x =$ ..... ° **[1]**

(c) A horizontal line $y = k$ intersects the graph of $y = \cos x$ at exactly two points in the interval $0° \leq x \leq 360°$.
The x-coordinate of one of these points is $242°$.
Find the value of k correct to 3 significant figures and find the x-coordinate of the second point of intersection.

$k =$ ..... , $x =$ ..... **[3]**

12 Algebraic techniques can be applied to solve trigonometric equations.

(a) Solve the equation for $0° \leq x \leq 360°$.

$$5\tan x + 14 = 0$$

Give your answers correct to 1 decimal place.

$x =$ ..... **[3]**

(b) Solve the equation for $0° \leq x \leq 360°$.

$$2\sin^2 x - \sin x = 0$$

$x =$ ..... **[4]**

1 $\sin x = \frac{1}{\sqrt{2}}$ **[M1]**

$x = 45°$ **[B1]**

$x = 135°$ **[B1]**

[Back to Question] *[Total Marks: 3]*

2 **(a)** $(180, -1)$ **[B1]**

(b) $60°$ **[B1]**

$300°$ **[B1]**

[Back to Question] *[Total Marks: 3]*

3 **(a)** Smooth sine wave shape passing through $(0, 0)$, $(180, 0)$ and $(360, 0)$ **[B1]**

Maximum turning point at $(90, 1)$ and minimum turning point at $(270, -1)$ **[B1]**

(b) $x = 30$ **[B1]**

$x = 150$ **[B1]**

[Back to Question] *[Total Marks: 4]*

4 **(a)** $\cos x = 0.5$ **[M1]**

$x = 60°$ **[B1]**

$x = 300°$ **[B1]**

(b) 0 (since $\sin x = \frac{5}{3} > 1$) **[B1]**

[Back to Question] *[Total Marks: 4]*

5 **(a)** Smooth U-shaped cosine curve starting at $(0, 1)$ and ending at $(4, 1)$ representing $360°$ **[B1]**

Curve passes through $(1, 0)$, $(3, 0)$ and has a minimum at $(2, -1)$ **[B1]**

(b) $(180°, -1)$ **[B1]**

(c) $\cos x = -0.5$ (or $-\frac{1}{2}$) **[M1]**

$x = 120°$ **[B1]**

$x = 240°$ **[B1]**

[Back to Question] *[Total Marks: 6]*

6 **(a)** Smooth sine curve passing through $(0, 0), (90°, 1), (180°, 0), (270°, -1), (360°, 0)$ **[B1]**

Smooth cosine curve passing through $(0, 1), (90°, 0), (180°, -1), (270°, 0), (360°, 1)$ **[B1]**

Intersection points shown approximately accurately at $x = 45°$ and $x = 225°$ **[B1]**

(b) $\tan x = 1$ **[M1]**

$x = 45°, 225°$ (Both required) **[A1]**

(c) Identifies the region between the two intersection points **[M1]**

$45° < x < 225°$ **[A1]**

[Back to Question] *[Total Marks: 7]*

7 **(a)** $\sin x = -\frac{\sqrt{3}}{2}$ **[M1]**

$240°$ **[A1]**

$300°$ **[A1]**

(b) $\cos^2 x = \frac{3}{4}$ **[M1]**

$\cos x = \pm\frac{\sqrt{3}}{2}$ **[M1]**

$30°$ and $150°$ **[A1]**

$210°$ and $330°$ **[A1]**

(Award maximum of 3 marks if $\pm$ is omitted, resulting in only two solutions)

[Back to Question] *[Total Marks: 7]*

8 **(a)** $\sin x = -0.75$ **[M1]**

Reference angle $= 48.6°$ (or $48.59...°$) **[M1]**

$x = 228.6°, 311.4°$ (Both required for the final mark) **[A1]**

(b) x-coordinate is $270°$ **[B1]**

y-coordinate is -1 **[B1]**

[Back to Question] *[Total Marks: 5]*

9 **(a)** Correct basic shape with multiple branches passing through $(0, 0)$, $(180, 0)$, and $(360, 0)$ **[B1]**

Asymptotes at $x = 90$ and $x = 270$ clearly indicated or graph showing asymptotic behaviour **[B1]**

(b) $\tan x = -\frac{5}{3}$ or $-1.66\ldots$ **[M1]**

$121.0°$ **[A1]**

$301.0°$ **[A1]**

[Back to Question] *[Total Marks: 5]*

10 **(a)** $\cos x = -0.25$ **[M1]**

$x = 104.5$ **[A1]**

$x = 360 - 104.47... = 255.5$ **[A1]**

(b) $\tan x = 2.5$ **[M1]**

$x = 68.2$ **[A1]**

$x = 68.19... + 180 = 248.2$ **[A1]**

[Back to Question] *[Total Marks: 6]*

11 **(a)** $\sin 30° = \frac{1}{2}$ leading to $a\left(\frac{1}{2}\right) - b = 0$ or $\frac{b}{a} = \frac{1}{2}$ **[M1]**

$a = 2$ and $b = 1$ **[A1]**

(b) 150 **[B1]**

(c) $k = \cos(242°) = -0.469$ **[B1]**

Uses symmetry of cosine curve: $360° - 242°$ **[M1]**

$x = 118°$ **[A1]**

[Back to Question] *[Total Marks: 6]*

12 **(a)** $\tan x = -2.8$ **[M1]**

$x = 109.7°$ (accept $109.6°$ to $109.7°$) **[B1]**

$x = 289.7°$ (accept $289.6°$ to $289.7°$) **[B1]**

(b) Factorises to $\sin x(2\sin x - 1) = 0$ **[M1]**

Identifies $\sin x = 0$ or $\sin x = 0.5$ (or $\frac{1}{2}$) **[M1]**

$x = 0°, 180°, 360°$ (all three correct for the mark) **[A1]**

$x = 30°, 150°$ (both correct for the mark) **[A1]**

[Back to Question] *[Total Marks: 7]*